Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

October 2026 · SolisCloud · Fleet review: 35 plants

All 35 solar plants Solar ESS monitors on SolisCloud · daily data 1–24 October 2026, 5-minute profiles for 24 October · site names removed

35	32.0 MWh	27.6 MWh	2.2 MWh	7.7 MWh
plants	**solar produced**	**bought from the grid**	**exported unpaid**	**delivered by batteries**
32 online · 1 fault · 2 offline	1–24 Oct	incl. one meter in doubt	at 10 plants	24 battery plants

What we found

- **Six plants need a site visit or a fault fixed now** – two batteries not working, about two-thirds of the panels not producing at a restaurant, a battery model error left for 101 days, solar not recorded at one home and one home offline since 14 September.
- **Settings waste energy at 13 plants:** unpaid export (about 2,000 kWh in 24 days), batteries held full or stopped early at night, grid charging at night, and charge/discharge limits far below what the battery can do.
- **Many batteries are full by 10:00–14:00**, after which the inverters throttle the panels. Moving daytime loads (water heating, pumps, air-con) to midday is the cheapest gain.
- **Six plants are good upgrade candidates** with high grid purchases – a resort, a commercial site, a nursing home and three homes.
- **Monitoring data needs tidying:** 22 of 24 battery plants have no battery size entered, two plants have the wrong PV capacity, two are listed on two platforms and one meter reads inconsistently.

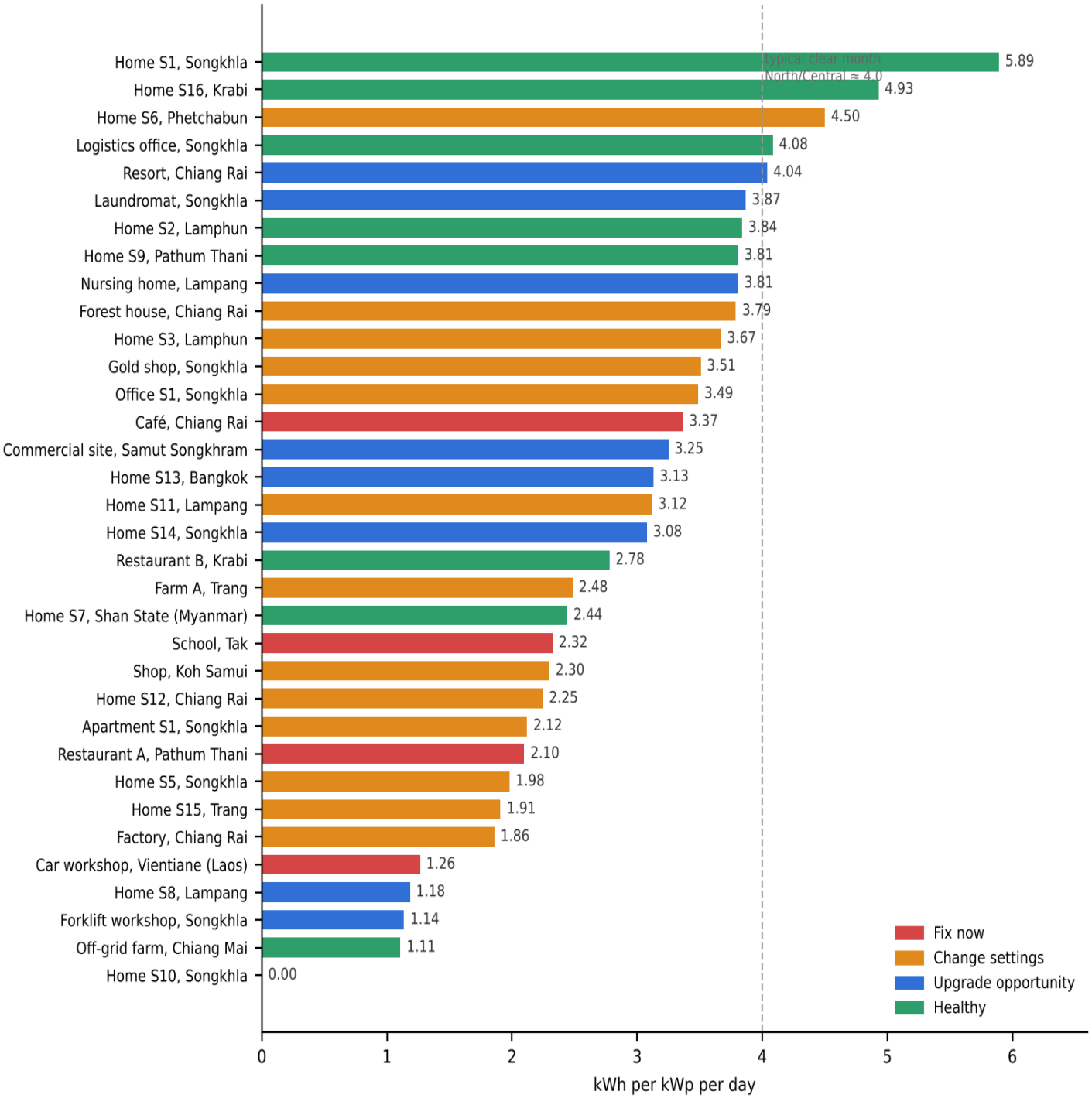
Priority list

Group	Plant	Action	Gain (estimate)
Fix now	**Restaurant A, Pathum Thani**	String check – about ⅔ of panels not producing	**+1,300–1,500 kWh/month**
Fix now	**School, Tak**	Battery BMS communication; reconnect datalogger	**+30–40 kWh/day self-use**
Fix now	**Car workshop, Vientiane (Laos)**	Battery not charging; 50 kWp throttled to load	**up to +100 kWh/day**
Fix now	**Café, Chiang Rai**	Select correct battery model; floor 20%	**−5–8 kWh/day grid**
Fix now	**Home S4, Chiang Rai**	Offline since 14 Sep – restore datalogger	**data**
Fix now	**Home S10, Songkhla**	Fix AC-coupled plant set-up (two platforms)	**data**
Change settings	**Forest house, Chiang Rai**	CT/load position, Self-Use, export 0 W, reserve SOC 20–30%	**≈ 700 kWh/month**
Change settings	**Home S6, Phetchabun**	Allow night discharge, raise charge limit, export 0 W	**≈ 350 kWh/month**
Change settings	**Farm A, Trang**	Grid over-voltage (56 trips) – PEA / cable check	**fewer trips**
Change settings	**Home S11, Lampang**	Turn off night grid charging	**−6 kWh/night**
Upgrade opportunity	**Commercial site, Samut Songkhram**	Second battery ≈ 20 kWh	**≈ −600 kWh/month grid**
Upgrade opportunity	**Resort, Chiang Rai**	Add ≈ 20 kWp on the 50 kW inverter; meter check	**≈ +2,400 kWh/month**

Gains are estimates from the 1–24 October data and the 24 October profiles. Southern plants (Songkhla, Trang, Krabi, Surat Thani) had a rainy October.

Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

Solar yield per kWp – every plant (1–24 Oct, kWh per kWp per day)



Home S4 (offline) is not shown. A low bar is not always a fault: plants with a full battery or no daytime load throttle their panels (see 'Change settings' and 'Upgrade'). Home S1 and Home S16 peak above their registered kWp, so their true yield per kWp is lower than shown.

Issues that repeat across the fleet

Issue	What we saw	What to do
Battery size not entered	22 of the 24 battery plants have no battery capacity entered in SolisCloud, so battery energy and health cannot be tracked.	Enter the battery kWh for every plant (Plant settings → battery capacity).
Unpaid export	About 2,000 kWh went to the grid in 24 days at 10 plants – Forest house, Chiang Rai 417, Home S6, Phetchabun 359, Apartment S1, Songkhla 225, Home S5, Songkhla 223, Factory, Chiang Rai 207, Office S1, Songkhla 163, Gold shop, Songkhla 156, Resort, Chiang Rai 115, Home S12, Chiang Rai 94, Home S15, Trang 81 kWh.	Without a PEA sell-back contract, set export limit 0 W (Solar Sell off) and charge the battery first.
Battery full by midday, panels throttled	At least 15 battery plants are full by 10:00–14:00 and the inverter then throttles the panels (e.g. Home S8, Lampang 10:00, Home S15, Trang 10:15, Commercial site, Samut Songkhram 12:45).	Run water heaters, pumps, air-con and EV charging 10:00–15:00; add battery where the evening grid use is high.
Low battery charge / discharge limits	Shop, Koh Samui (charge 2.2 kW), Home S3, Lamphun (discharge 1.6 kW), Farm A, Trang (3.2 kW), Home S8, Lampang (1.5 kW).	Raise the charge/discharge current limits up to the battery BMS rating.
Batteries not working	School, Tak and Car workshop, Vientiane (Laos) show 100% all day with no charging; Café, Chiang Rai has a battery model error.	Site check of BMS cable, battery breaker and battery model selection.
Wrong plant set-up	Home S10, Songkhla records no solar (AC-coupled to a Huawei inverter); Home S1, Songkhla and Home S16, Krabi peak above their registered kWp; Laundromat, Songkhla and Home S10, Songkhla are listed twice (Solis + Huawei).	Correct PV capacity and plant type so monthly reports are right.
Grid voltage trips	Farm A, Trang 56 over-voltage trips, Forklift workshop, Songkhla 17 under-voltage, Home S13, Bangkok 10 over-voltage, Home S1, Songkhla 8 under-voltage.	Measure voltage at the meter; ask PEA to adjust the transformer tap; check AC cable sizes.
Rainy south	Songkhla, Trang, Krabi and Surat Thani had a wet October; 2–3 kWh/kWp/day is normal there this month.	Compare southern plants with each other, not with the north.

Recommended standard settings for hybrid plants without a sell-back contract

Work mode	Self-Use (not Backup/Reserve unless the owner needs outage backup)
Export limit	On, 0 W (Solar Sell off) – the battery charges first, then panels follow the load
Battery low / overdischarge SOC	20% (LFP); reserve SOC 20–30% only where outage backup is wanted
Battery charge / discharge current	Up to the battery BMS rating – not the 25–50 A default
Grid charging	Off, unless on a TOU tariff with a cheap night rate
Time-of-Use slots	Charge 00:00–24:00 from solar; discharge allowed 00:00–24:00
Battery model	The exact model installed – a wrong model limits the battery
Plant settings	Correct PV kWp, battery kWh and plant type (AC-coupled where applicable)

Always check the battery datasheet / BMS limits before raising current, and confirm with the owner before changing backup reserve.

Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

FIX NOW · 6 plants

Home S4, Chiang Rai

Offline since 14 Sep

17.76 kWp · 10 kW 1-phase hybrid inverter · no battery

–	–	–	–	–	–
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery

No data on 24 Oct

No data in October

24 October, 5-minute data

What we found

• No data for six weeks: last upload 14 Sep 2026, so October production is unknown.
• 17.76 kWp of panels on a 10 kW inverter (DC/AC 1.78) – heavy clipping at midday is likely even when it is online.

Daily totals, 1–24 October

What to do

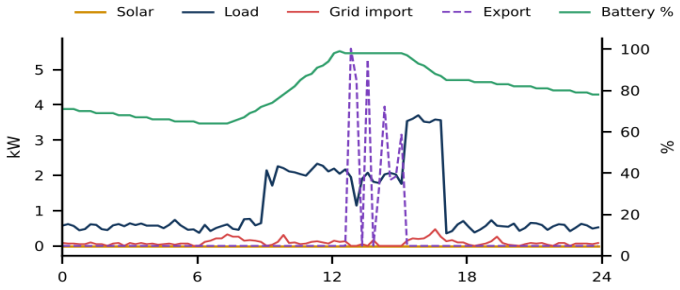
• Site visit or phone call: check the inverter is on and the WiFi/4G datalogger is powered and connected; re-pair it to the router if the WiFi password changed.
• Once online, check midday output against the 10 kW limit; consider moving part of the array to a second inverter.

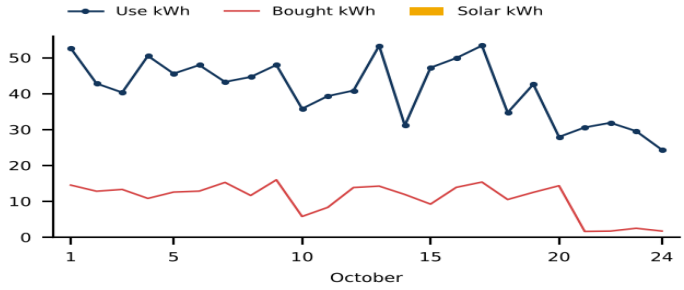
Home S10, Songkhla

Solar not recorded on SolisCloud

10 kWp · 16 kW 1-phase hybrid inverter · battery

0	0.00	74%	10.7	30	15.3
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery





24 October, 5-minute data

What we found

• SolisCloud shows 0 kWh solar all month, but the battery still charges 5 kW at midday and the meter sees power coming in.
• The panels appear to be on a separate inverter on another monitoring platform (listed there with 0 kWp), AC-coupled to this hybrid inverter.
• Two battery over-voltage alarms (OV-Vbatt) in October.

Daily totals, 1–24 October

What to do

• Fix the plant set-up: on SolisCloud mark the 10 kWp as AC-coupled (or set PV capacity to 0) and enter the real 10 kWp on the Huawei plant, so each platform reports correctly.
• Check battery charge voltage settings after the OV-Vbatt alarms.

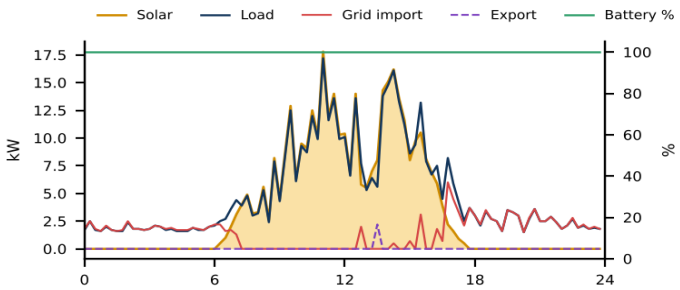
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

Car workshop, Vientiane (Laos)

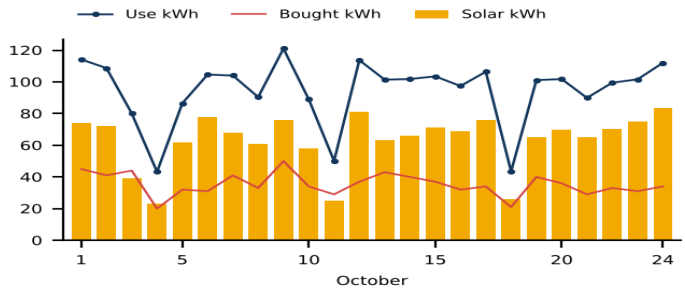
Battery idle · 50 kWp at 1.3 kWh/kWp

50 kWp · 50 kW 3-phase hybrid inverter · battery

1517	1.26	63%	35.3	14	0.0
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Lowest real yield in the fleet: 1.26 kWh/kWp/day (63 kWh/day from 50 kWp). Peak only 17.8 kW on a clear day.
• The battery shows 100% all day and never charges or discharges.
• Solar follows the workshop load (zero export), so most of the array is switched off at midday; at night the site buys 2–6 kW.

What to do

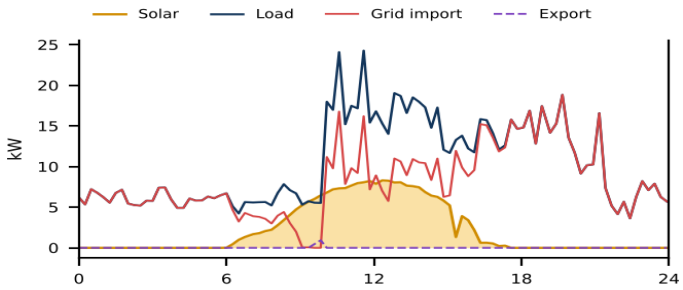
• Check whether the battery is actually connected and communicating (BMS cable, battery breaker, battery model setting).
• If the battery cannot work, either fix it or remove it from the configuration; then size the system for the load.
• Ask the owner whether export to the grid (EDL) is allowed – if yes, set an export limit instead of zero export.
• Potential: 100+ kWh/day more solar if storage or export is available.

Restaurant A, Pathum Thani

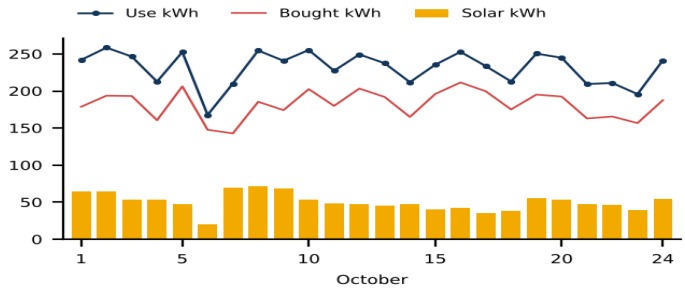
About ⅔ of panels not producing

24 kWp · 20 kW 3-phase hybrid inverter · no battery

1208	2.10	21%	182.1	22	0.0
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• 24 kWp but only 8.3 kW peak on 24 Oct (0.35 kW per kWp) while the restaurant was importing 15–18 kW – nothing should be limiting the panels.
• Yield 2.1 kWh/kWp/day against 3.8–4.0 at similar central/northern sites.
• Solar covers only 21% of a very large load (231 kWh/day).

What to do

• Site check of every string: DC isolators, fuses, MC4 connectors and MPPT input currents; one or two strings are probably disconnected or tripped.
• Expected gain once fixed: about 1,300–1,500 kWh/month.

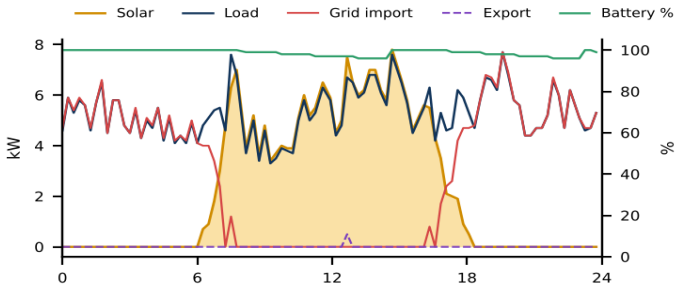
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

School, Tak

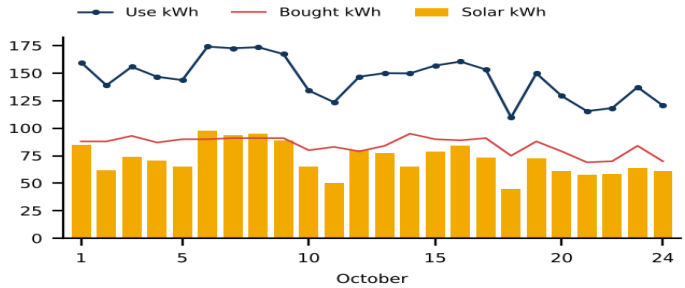
Battery not working · offline

30.96 kWp · 50 kW 3-phase hybrid inverter · no battery

1726	2.32	42%	84.8	13	0.0
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Went offline at 01:01 on 25 Oct.
• Battery reads 96–100% all day and never charges or discharges; 34 battery-management (BMS) alarms in October.
• Without the battery the inverter limits solar to the school's load: only 7.8 kW peak from 31 kWp, 2.3 kWh/kWp/day.
• Buys about 85 kWh a day from the grid.

What to do

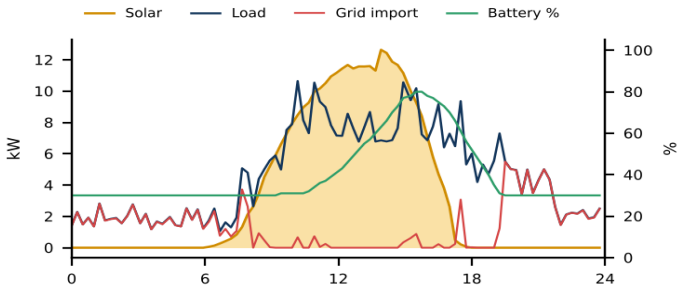
• Restore the datalogger connection.
• Check the battery: BMS communication cable (CAN/RS485), battery breaker and the battery model selected in the inverter; clear the BMS alarms.
• With the battery working, midday surplus can be stored for the afternoon – expected 30–40 kWh/day more solar used.

Café, Chiang Rai

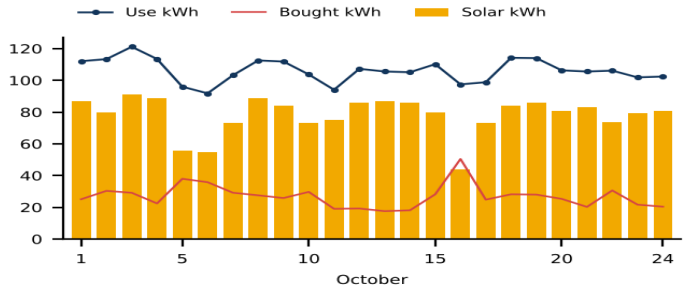
Battery model error for 101 days

23.22 kWp · 15 kW 3-phase hybrid inverter · battery

1876	3.37	75%	26.9	0	19.8
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Active alarm 'BattChoose-Err' since 15 Jul 2026: the battery model selected in the inverter does not match the battery.
• The battery only works between 30% and 80% and never reaches full, so evening power is bought from the grid (27 kWh/day).
• 23.2 kWp on a 15 kW inverter – peak 12.6 kW, so clipping is small.

What to do

• Select the correct battery model in the inverter and restart it (as the alarm advises).
• Then set the battery low/overdischarge SOC to 20% so the full battery is used.
• Expected: 5–8 kWh/day less grid.

Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

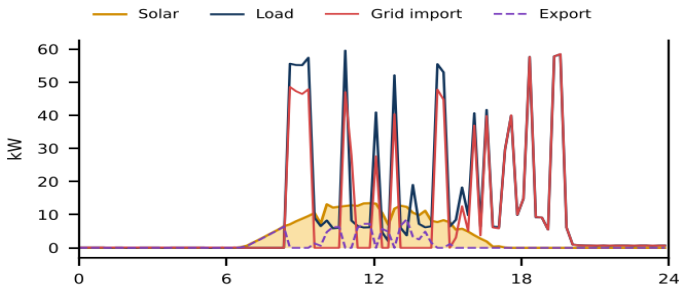
CHANGE SETTINGS · 13 plants

Factory, Chiang Rai

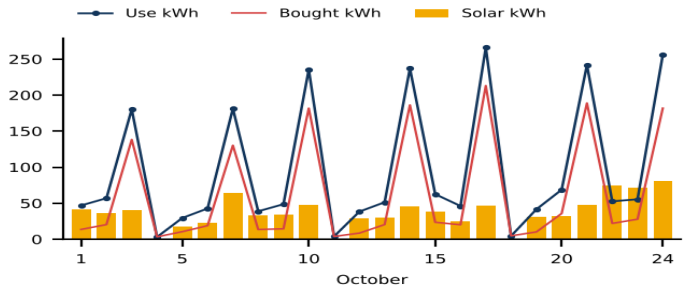
Meter readings · export on closed days

20 kWp · 20 kW 3-phase hybrid inverter · no battery

892	1.86	35%	62.2	207	0.0
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Load readings jump to 55 kW for single samples and grid readings swing ±48 kW – meter/CT readings are unreliable.
• 207 kWh exported; production near zero on Sundays (4, 11, 18 Oct) when the factory is closed.

What to do

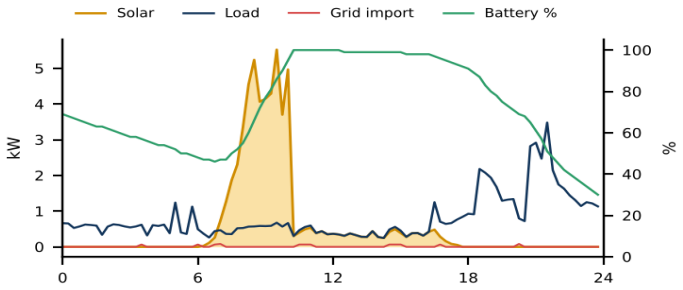
• Finish the CT/meter check and confirm zero export works.
• Balance the load across the three phases (being monitored separately).

Home S15, Trang

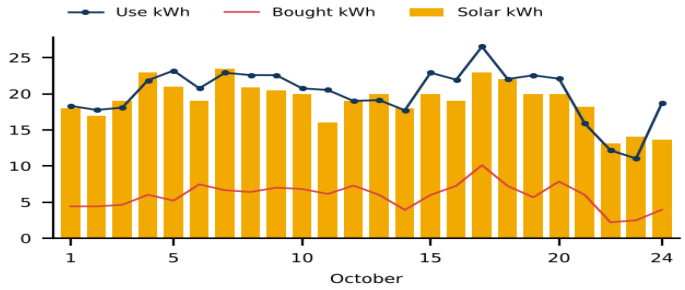
Battery full by 10:15 · curtailed

10 kWp · 10 kW 3-phase hybrid inverter · battery

458	1.91	70%	5.9	81	10.0
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Battery is full by 10:15, after which the panels are throttled to the house load (0.3–0.5 kW).
• Only 1.9 kWh/kWp/day; 81 kWh exported unpaid.

What to do

• Move daytime loads (water heater, washing, air-con, pump) to 10:00–15:00.
• Set export limit to 0 W, or consider a larger battery if evening use grows.

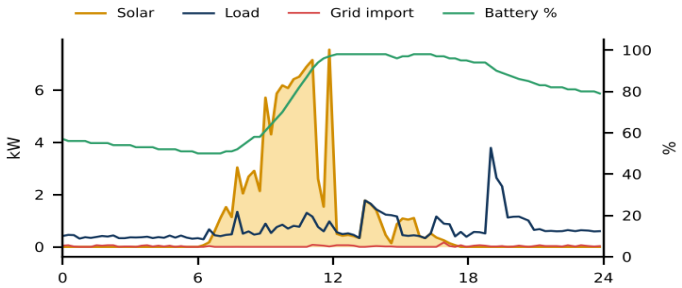
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

Home S5, Songkhla

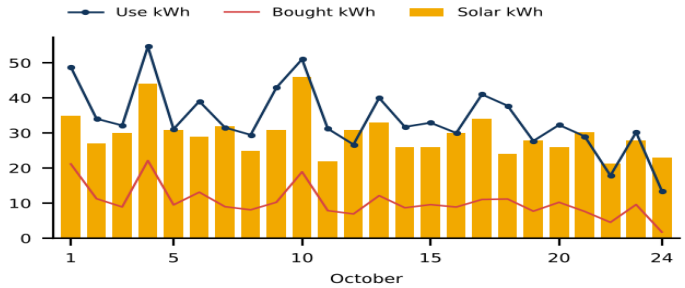
Battery only used 50–98% · export

15 kWp · 15 kW 3-phase hybrid inverter · battery

713	1.98	69%	10.4	223	21.3
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

- Battery never goes below 50% – about half of it is never used.
- Sells 223 kWh to the grid while buying 251 kWh (unpaid export).
- Battery full by noon, then solar is throttled; 29 alarms.

What to do

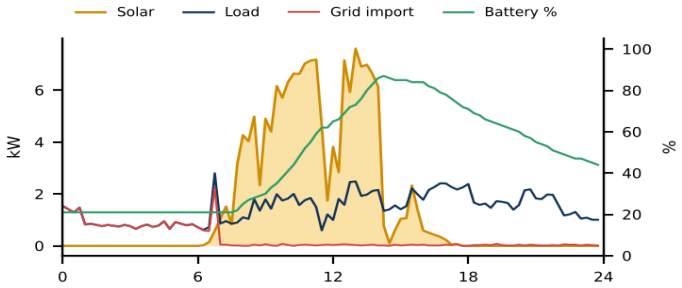
- Lower battery low/overdischarge SOC to 20%.
- Set export limit to 0 W (Solar Sell off) unless the owner has a sell-back contract.
- Expected: 6–9 kWh/day less grid.

Apartment S1, Songkhla

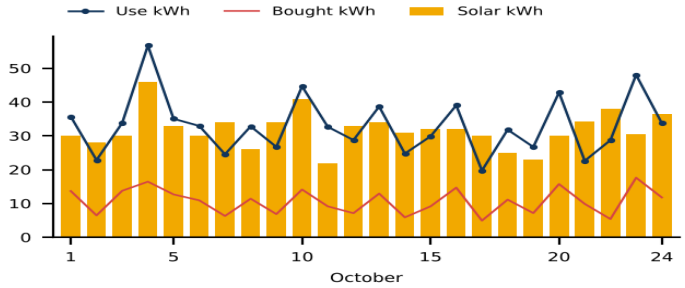
225 kWh exported

15 kWp · 15 kW 3-phase hybrid inverter · battery

763	2.12	68%	10.6	225	19.4
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

- Sells 225 kWh while buying 255 kWh (unpaid export).
- Battery cycles 21–87%; yield 2.1 kWh/kWp/day (rainy southern October).

What to do

- Set export limit to 0 W / Solar Sell off, so surplus goes to the battery first.
- Check the Time-of-Use charge slots allow charging all day.

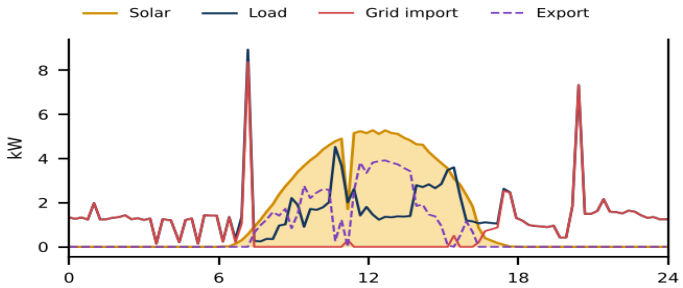
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

Home S12, Chiang Rai

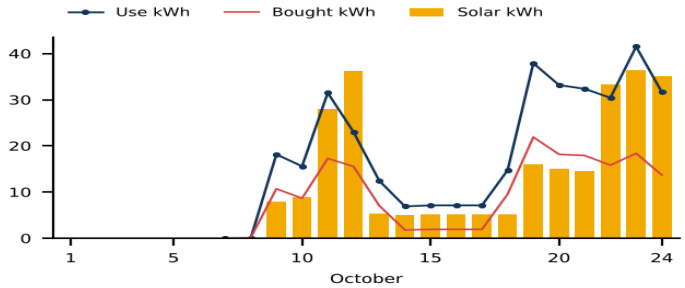
New · repeated data · export

6.5 kWp · 10 kW hybrid inverter · no battery

263	2.25	48%	10.1	94	0.0
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Connected 7 Oct. 13–17 Oct show exactly the same 5.2 kWh every day – the datalogger was probably not updating.
• No battery; exports up to 3.9 kW at midday (94 kWh so far).
• 6.5 kWp on a 10 kW inverter – room for about 4 kWp more panels.

What to do

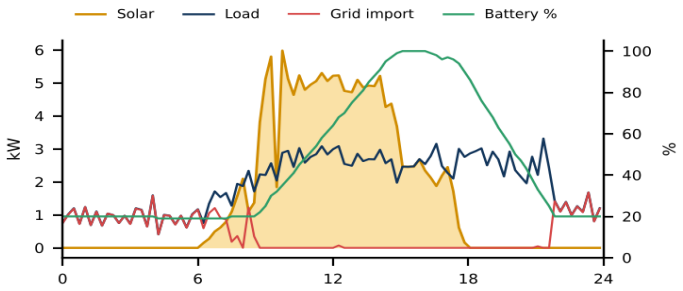
• Confirm the datalogger now reports live data.
• Set export limit to 0 W unless a PEA permit is in place; consider more panels or a battery later.

Shop, Koh Samui

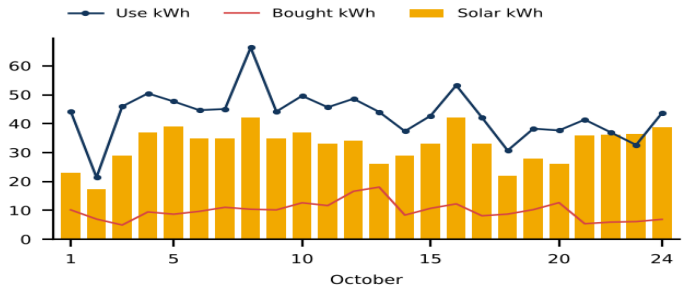
Battery charge rate capped

14.19 kWp · 16 kW 1-phase hybrid inverter · battery

783	2.30	77%	9.8	0	10.8
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Battery charges at a flat 2.2 kW for three hours, so the inverter holds the panels at 5–6 kW from 14.2 kWp.
• Battery only full at 14:50; yield 2.3 kWh/kWp/day.

What to do

• Raise the battery charge current limit (e.g. 50 → 100 A, within the BMS limit) so the battery fills earlier from solar.
• Expected: 4–6 kWh/day more solar used.

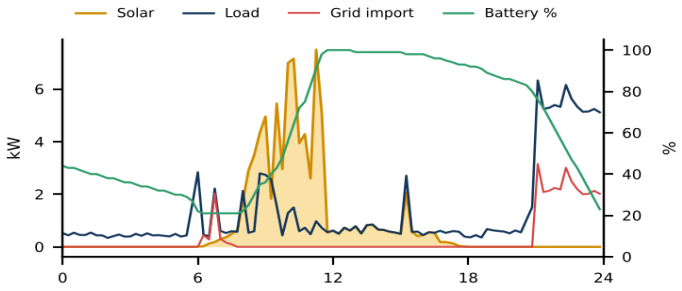
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

Farm A, Trang

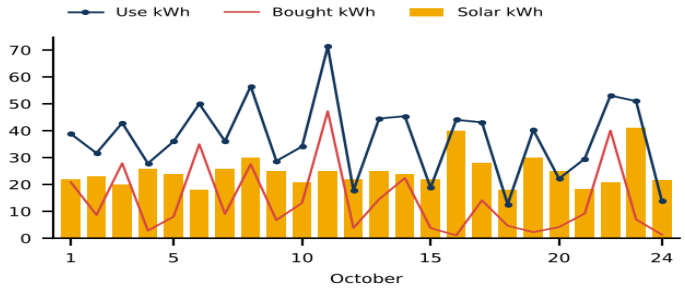
56 grid over-voltage trips

10 kWp · 10 kW 1-phase hybrid inverter · battery

596	2.48	62%	14.0	0	11.3
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• 56 'OV-G-V01' grid over-voltage alarms in October – the inverter disconnects each time and production stops.
• Battery discharge is capped at about 3.2 kW while the evening load is 5 kW, so the grid supplies the rest.
• Battery full by 11:30, then the panels are throttled for the rest of the day.

What to do

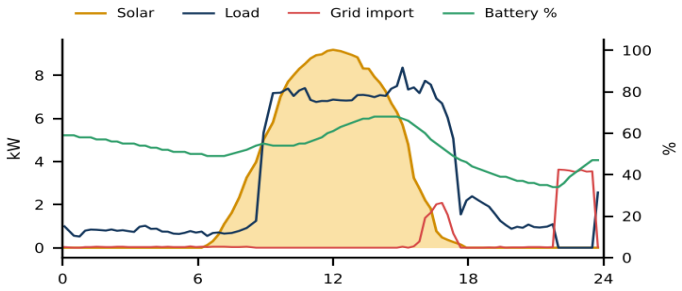
• Measure grid voltage at the meter; ask PEA to check the transformer tap if it is above 240–250 V. Check AC cable size/voltage drop to the inverter.
• Raise the battery discharge current limit (within the BMS limit).
• Move pumps and other big loads to 11:00–15:00; a second battery would also use the afternoon surplus.

Home S11, Lampang

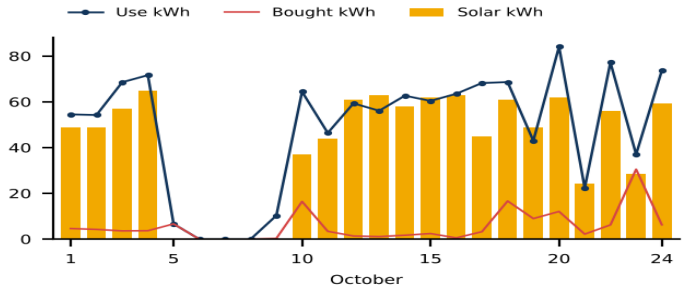
Grid charging at night · 5 days down

13.265 kWp · 8 kW 1-phase hybrid inverter · battery

993	3.12	88%	5.7	0	15.8
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Battery charges from the grid at about 4.2 kW from 22:00 to 23:30 (≈6 kWh a night bought just to fill the battery).
• No production 5–9 Oct (5 days).
• 13.3 kWp on an 8 kW inverter (DC/AC 1.66).

What to do

• Turn off grid charging / the night charge time slot unless the owner is on a TOU tariff and wants it.
• Find out why the system stopped 5–9 Oct (inverter fault, breaker, or power cut) and enable fault notifications.
• Watch midday clipping; the extra panels mainly help mornings and afternoons.

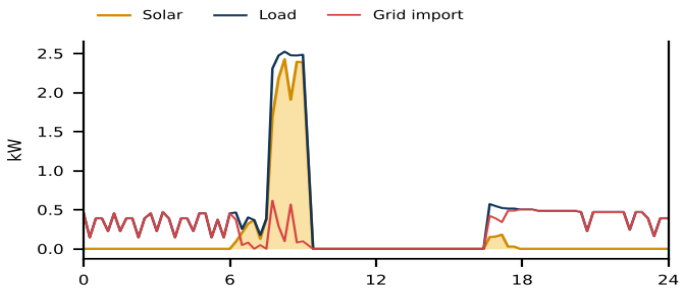
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

Office S1, Songkhla

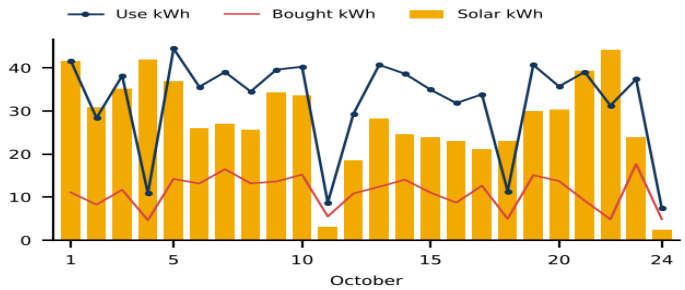
Data gaps · 163 kWh exported

8 kWp · 8 kW 1-phase hybrid inverter · no battery

670	3.49	65%	11.2	163	0.0
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Data missing for hours (e.g. 09:15–15:30 on 24 Oct, and 11 Oct) – the datalogger drops out.
• No battery; 163 kWh exported unpaid.

What to do

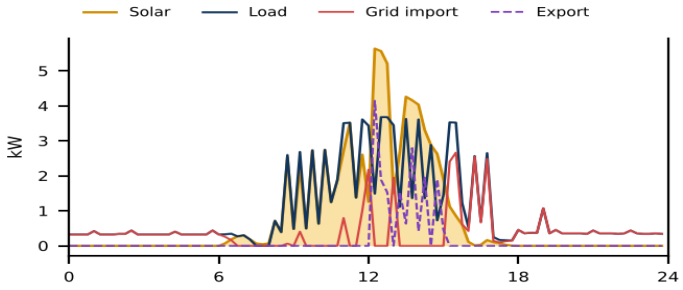
• Check the WiFi signal at the datalogger; move the router or add a repeater.
• Set export limit to 0 W, or add a battery to use the surplus in the evening.

Gold shop, Songkhla

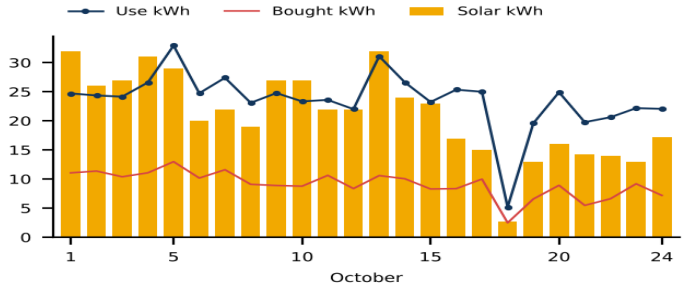
Export · zero-export from 16 Oct

6 kWp · 6 kW 1-phase hybrid inverter · no battery

505	3.51	62%	9.1	156	0.0
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• No battery. 156 kWh exported before 16 Oct; from 16 Oct export stops but daily solar also drops from 20–30 to 13–17 kWh – the panels are now throttled to the shop load.

What to do

• A 10 kWh battery would store about 8–10 kWh/day that is now lost.
• Otherwise move shop loads (air-con, polishing) to the sunniest hours.

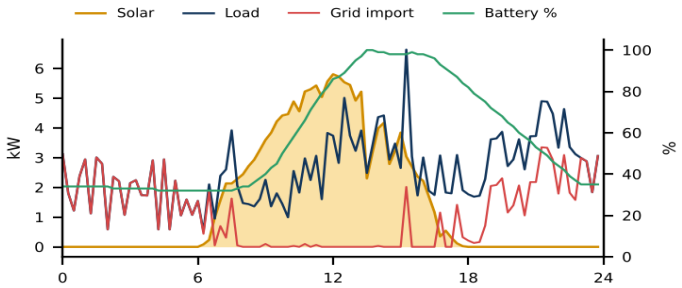
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

Home S3, Lamphun

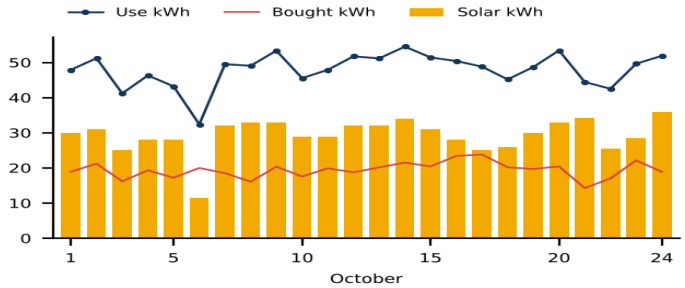
Discharge capped 1.6 kW · floor 35%

8 kWp · 8 kW 1-phase hybrid inverter · battery

705	3.67	60%	19.4	0	10.3
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• At night the battery discharges at only 1.6 kW while the house uses 3–5 kW; the grid supplies the rest (19 kWh/day bought).
• Battery stops at about 35% in the morning.
• Battery full by 13:30, then solar is throttled.

What to do

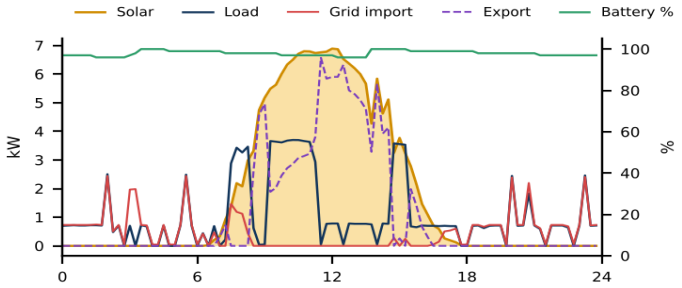
• Raise the battery discharge current limit and lower the battery low SOC to 20%.
• Expected: 4–6 kWh/day less grid.

Forest house, Chiang Rai

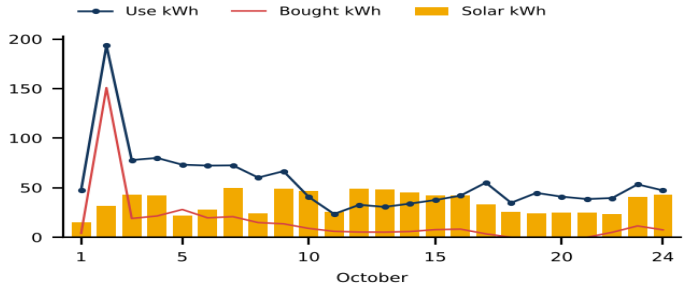
Battery reserved · 417 kWh exported

9.28 kWp · 8 kW 1-phase hybrid inverter · battery

844	3.79	52%	15.3	417	3.7
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Battery sits at 96–100% and is not used: the house buys power at night while the battery is full.
• All house load is on the backup port, and the grid CT does not see it – the inverter exports up to 6.6 kW at midday: 417 kWh sent to the grid in 24 days, unpaid.
• 132 alarms in the period, mostly NO-Grid.

What to do

• Move the house load (or the CT) so the meter measures the whole house; set Work Mode to Self-Use and export limit to 0 W.
• Lower the backup/reserve SOC to 20–30% so the battery covers the night.
• Expected: about 400 kWh/month less export and 300 kWh/month less grid purchase.

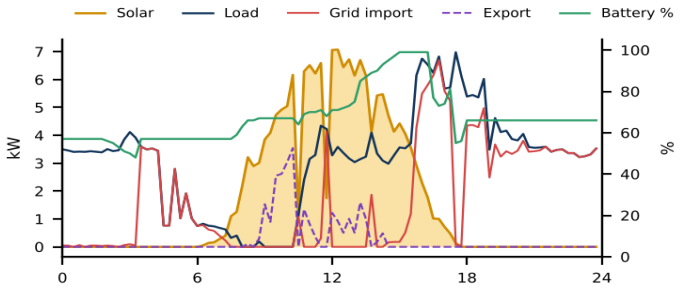
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

Home S6, Phetchabun

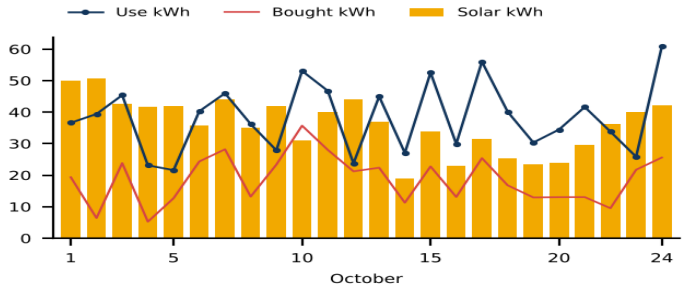
Battery stops at night · 359 kWh exported

8 kWp · 8 kW 1-phase hybrid inverter · battery

864	4.50	51%	18.7	359	12.5
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• At night the battery stops discharging with 66% left while the house buys 3.4 kW from the grid.
• At midday the battery charges at only about 3 kW and the rest (up to 3.5 kW) is exported: 359 kWh in 24 days, unpaid.
• Good yield (4.5 kWh/kWp/day).

What to do

• Check Time-of-Use / discharge time slots: allow discharge 00:00–24:00 down to 20%.
• Raise the battery charge current limit (within the battery's BMS limit) and set the export limit to 0 W unless the owner has a PEA sell-back contract.
• Expected: about 350 kWh/month more self-use.

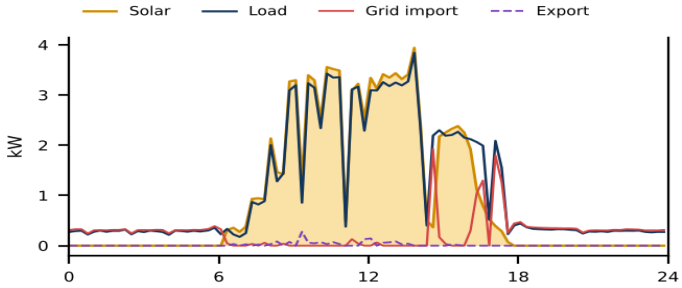
UPGRADE OPPORTUNITY · 8 plants

Forklift workshop, Songkhla

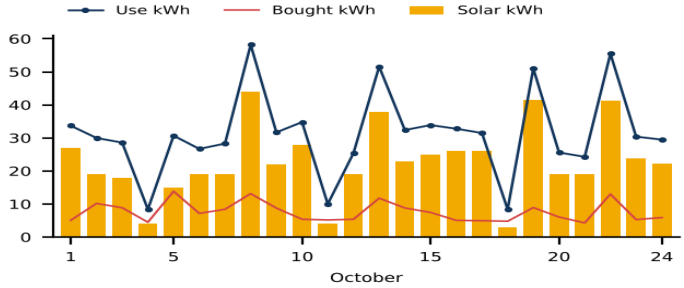
Panels throttled · no daytime load

20 kWp · 20 kW 3-phase hybrid inverter · no battery

546	1.14	76%	7.6	14	0.0
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Lowest yield on a 20 kWp system: 1.14 kWh/kWp/day. No battery and zero export, so the inverter only produces what the site uses (3.9 kW peak).
• Site uses only 31 kWh/day; 17 grid under-voltage alarms.

What to do

• Charge forklifts and run compressors between 10:00 and 15:00 to use the free solar.
• A battery is not worth it at today's low grid purchase (8 kWh/day). Check grid voltage because of the under-voltage trips.

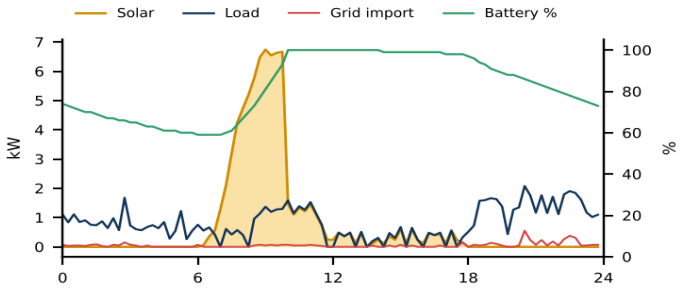
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

Home S8, Lampang

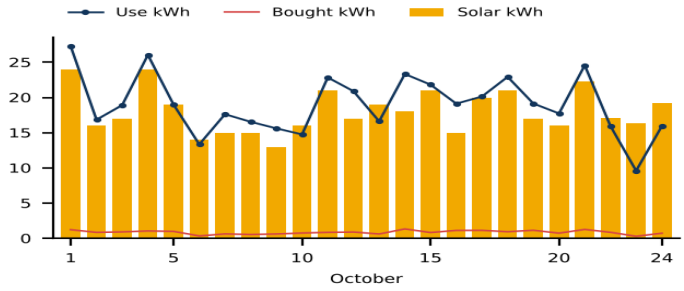
Not dirty – just full

15.24 kWp · 16 kW 1-phase hybrid inverter · battery

433	1.18	96%	0.9	0	10.7
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Low yield (1.18 kWh/kWp/day) is NOT dirty panels: the battery is full by 10:00 and the inverter then throttles the panels to the 0.5–1.5 kW load.
• Buys almost nothing from the grid (21 kWh in 24 days). Battery discharge capped at 1.5 kW.

What to do

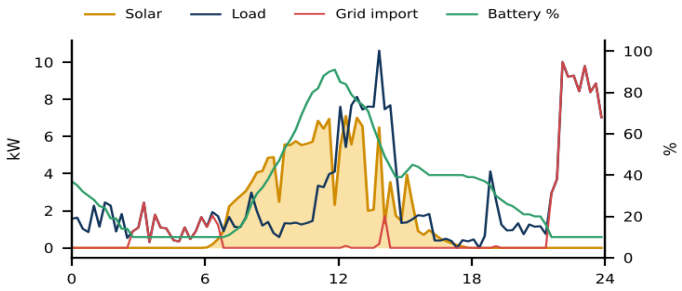
• Panel cleaning will not raise output – no need.
• Use the surplus: electric water heating, pumping, air-con or EV charging in the daytime.

Home S14, Songkhla

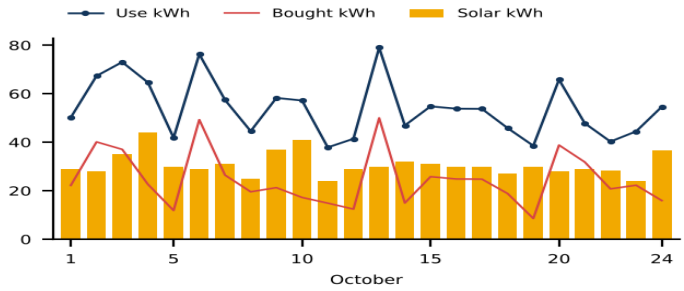
Battery too small for evenings

10 kWp · 10 kW 1-phase hybrid inverter · battery

739	3.08	54%	24.7	0	12.0
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Battery reaches its 10% floor by about 21:30; evening load 9–10 kW then comes from the grid (25 kWh/day bought).
• Battery low SOC set to 10% – very deep for daily cycling.

What to do

• Add 10–15 kWh of battery.
• Raise the battery low SOC to 15–20% to protect battery life.

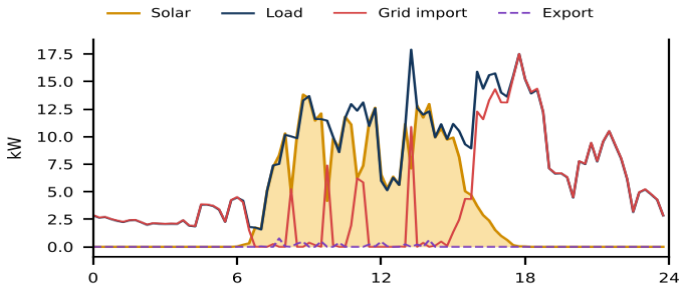
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

Home S13, Bangkok

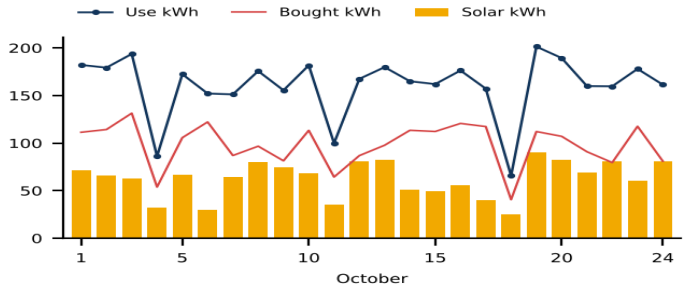
Add battery 20–30 kWh

20 kWp · 20 kW 3-phase hybrid inverter · no battery

1503	3.13	39%	98.4	19	0.0
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• No battery on a hybrid inverter. Evening load 10–17 kW, buying 98 kWh/day.
• Solar 3.1 kWh/kWp/day (Bangkok). 10 grid over-voltage trips in October.

What to do

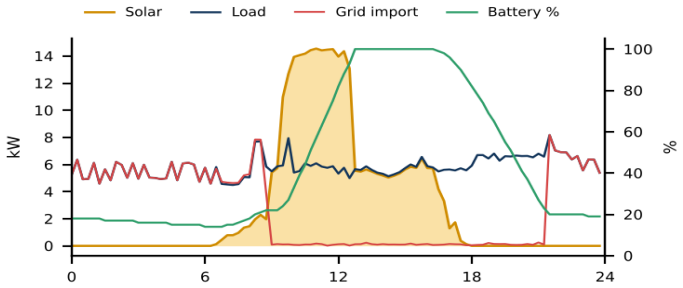
• Add 20–30 kWh of battery: about 600–900 kWh/month less grid.
• Measure grid voltage at the meter because of the over-voltage trips.

Commercial site, Samut Songkhram

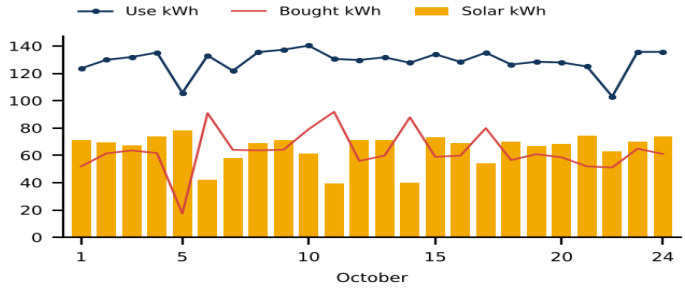
Second battery: ≈ +600 kWh/month

20.079 kWp · 20 kW hybrid inverter · battery

1566	3.25	51%	63.2	1	26.2
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Battery full by 12:45; the panels then drop from 14.5 kW to 5.5 kW for three hours (about 20 kWh/day thrown away).
• Battery empty by 21:30; the site then buys 6–7 kW all night (63 kWh/day).
• Yield 3.25 kWh/kWp/day.

What to do

• Add a second battery of about 20 kWh – it would absorb the afternoon surplus and cover the late evening: about 600 kWh/month less grid.
• Keep battery low SOC at 20%.

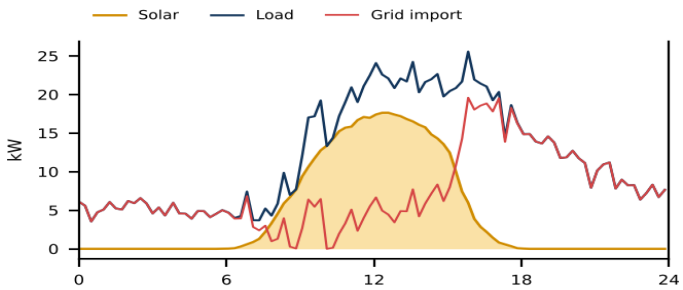
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

Nursing home, Lampang

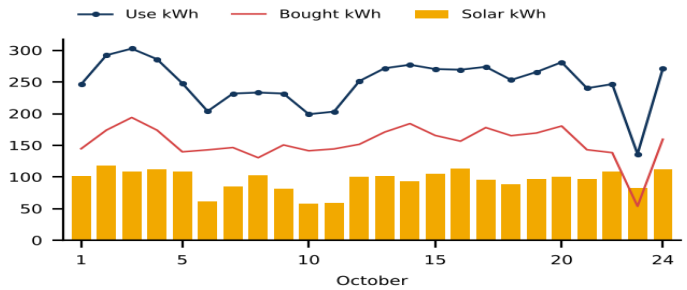
Add PV + battery

25.08 kWp · 20 kW 3-phase hybrid inverter · no battery

2291	3.81	38%	154.1	3	0.0
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Solar is fully used (good yield 3.8 kWh/kWp/day) but covers only 38% of the load (249 kWh/day).
• 25 kWp on a 20 kW inverter, no battery; the home buys 154 kWh/day, including 7–14 kW at night.

What to do

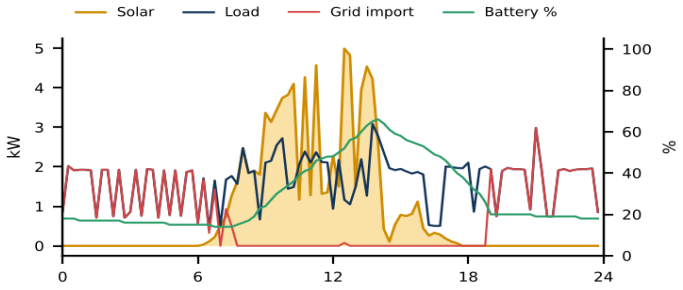
• Add a second inverter with 20–25 kWp more panels (+2,000–2,500 kWh/month).
• Add a battery for the night load.

Laundromat, Songkhla

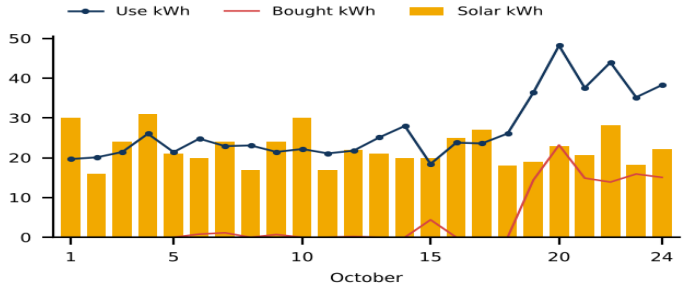
Add 4 kWp panels

5.8 kWp · 10 kW 3-phase hybrid inverter · battery

538	3.87	84%	4.4	46	11.5
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• 5.8 kWp on a 10 kW inverter (DC/AC 0.58); battery never gets above 66%.
• Buys 3 kW at night; 84% solar share.
• Also listed on Huawei FusionSolar with 0 kWp – duplicate plant.

What to do

• Add about 4 kWp of panels (inverter has room): about +450 kWh/month.
• Remove or merge the duplicate Huawei listing.

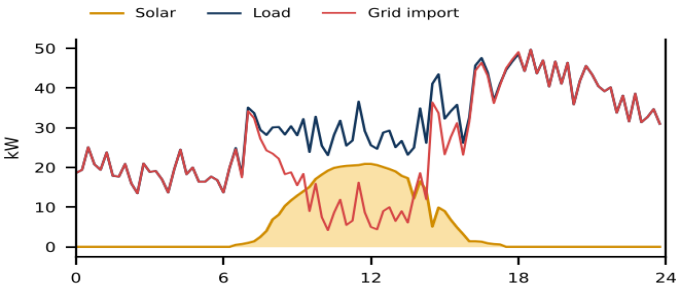
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

Resort, Chiang Rai

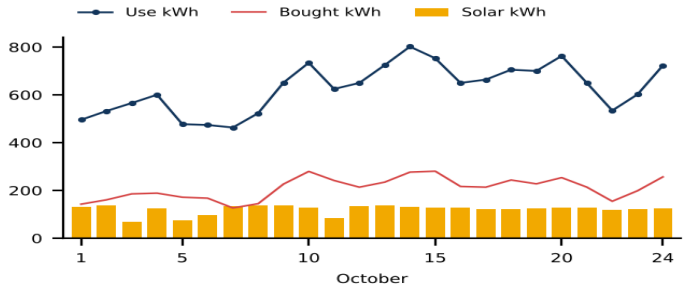
Add 20 kWp · meter check

30.03 kWp · 50 kW 3-phase hybrid inverter · no battery

2911	4.04	67%	209.5	115	0.0
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Very large load: 628 kWh/day. Solar 30 kWp gives 4.0 kWh/kWp/day (good) but covers only a small part.
• Meter figures do not add up: the 24 Oct profile shows 30–49 kW imported overnight, far more than the 210 kWh/day daily import figure.
• 50 kW inverter with only 30 kWp of panels.

What to do

• Check the grid meter / CT ratio setting so import and load are reported correctly.
• Add about 20 kWp of panels on the existing 50 kW inverter: about +2,400 kWh/month.
• Battery storage for the evening peak is the next step.

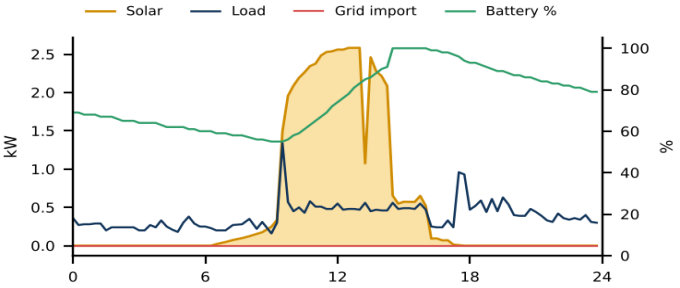
HEALTHY · 8 plants

Off-grid farm, Chiang Mai

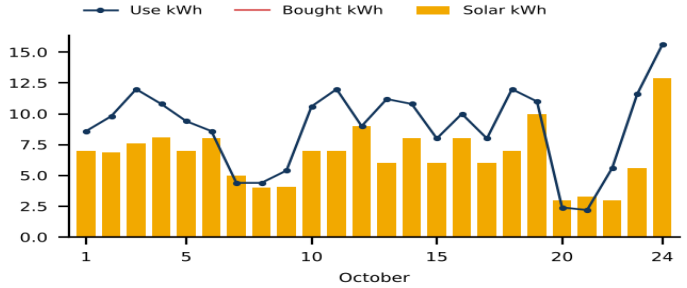
Off-grid · oversized for load

6 kWp · 6 kW hybrid inverter · battery

159	1.11	100%	0.0	0	3.8
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Off-grid: all load (≈4 kWh/day) on the backup port; battery full by 14:30.
• Low yield (1.1 kWh/kWp/day) only because there is little to power.

What to do

• No action; extra daytime loads can be added freely.

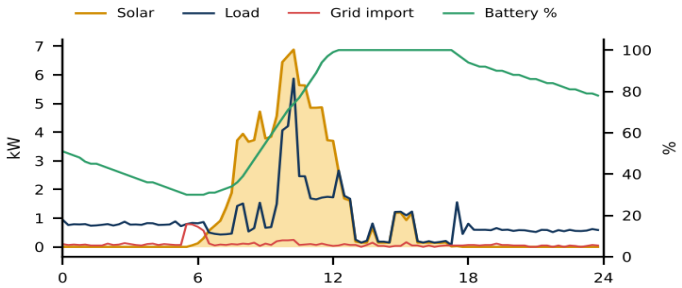
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

Home S7, Shan State (Myanmar)

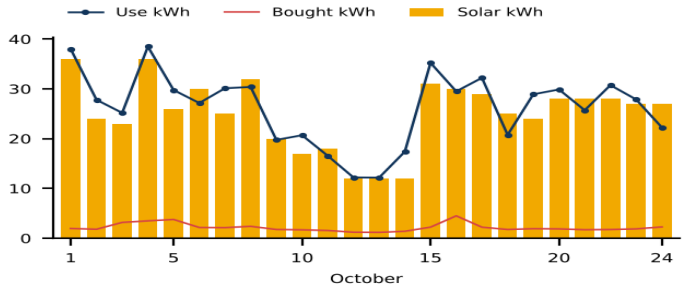
92% solar

10.24 kWp · 8 kW 1-phase hybrid inverter · battery

600	2.44	92%	2.2	0	11.2
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• 92% solar share. Time zone UTC+06:30 is correct for Shan State, Myanmar.
• Battery full by 12:00, then solar throttled; 10.24 kWp on 8 kW inverter.

What to do

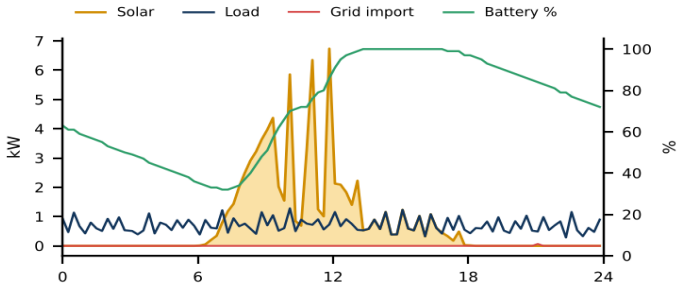
• No action; daytime loads could use the afternoon surplus.

Restaurant B, Krabi

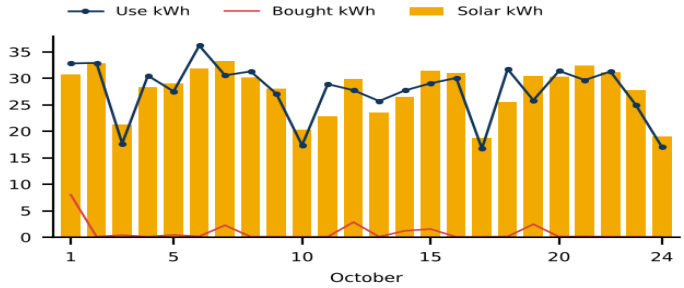
97% solar

10 kWp · 10 kW 1-phase hybrid inverter · battery

667	2.78	97%	0.9	0	11.0
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• 97% solar share; battery full by 12:50, then solar is throttled.

What to do

• No action. Run extra loads at midday if needed.

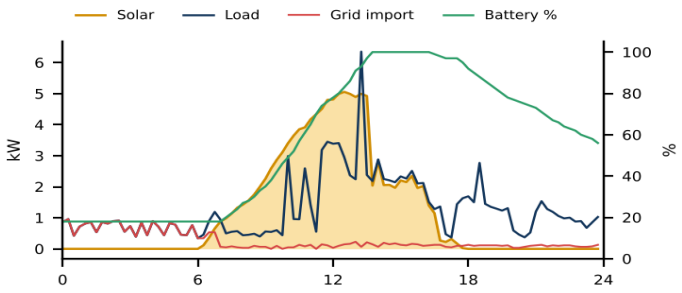
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

Home S9, Pathum Thani

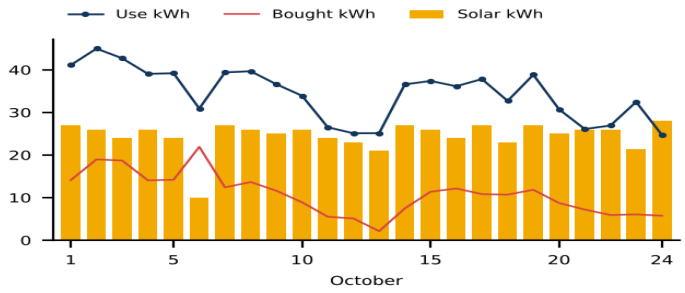
Healthy

6.45 kWp · 8 kW hybrid inverter · battery

589	3.81	69%	10.8	0	10.5
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• 3.8 kWh/kWp/day; 69% solar share; battery full by 13:45 and empty before dawn.

What to do

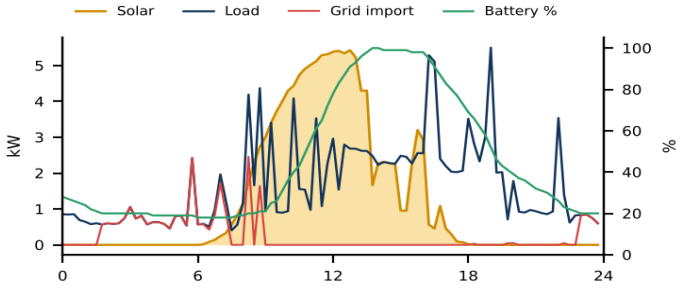
• A small extra battery (5 kWh) would cover the early-morning grid use.

Home S2, Lamphun

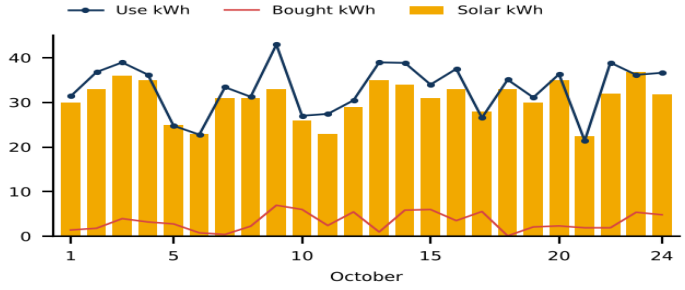
90% solar

8 kWp · 8 kW 1-phase hybrid inverter · battery

737	3.84	90%	3.2	0	12.4
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• 90% solar share; 3.8 kWh/kWp/day; battery full by 13:30.

What to do

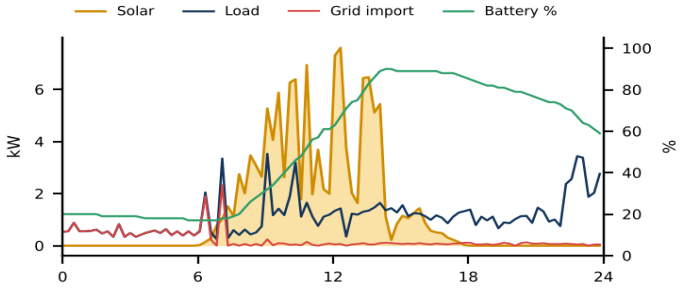
• No action.

Logistics office, Songkhla

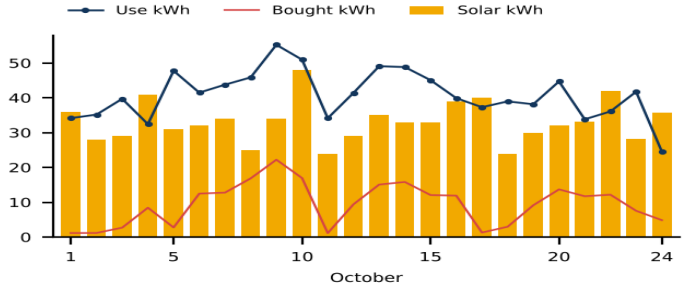
Healthy

8.12 kWp · 8 kW 1-phase hybrid inverter · battery

796	4.08	77%	9.4	0	15.1
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• 4.1 kWh/kWp/day (south); 77% solar share; battery 17–90%.

What to do

• No action.

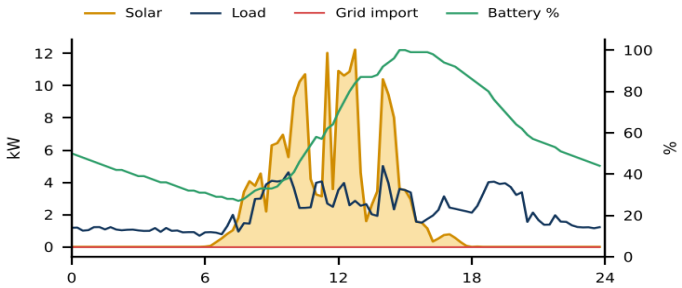
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang
Chiang Rai 57100, Thailand
Tel 063 151 6694, 089 632 5829
www.solaress.org · LINE: solaress1

Home S16, Krabi

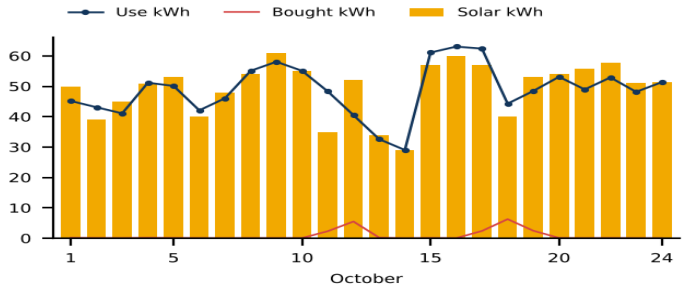
98% solar · check kWp

10 kWp · 10 kW 1-phase hybrid inverter · battery

1183	4.93	98%	0.9	0	22.7
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Runs almost fully on solar (98%); grid nearly unused.
• Peak 12.2 kW on '10 kWp' – the registered capacity is probably too low.

What to do

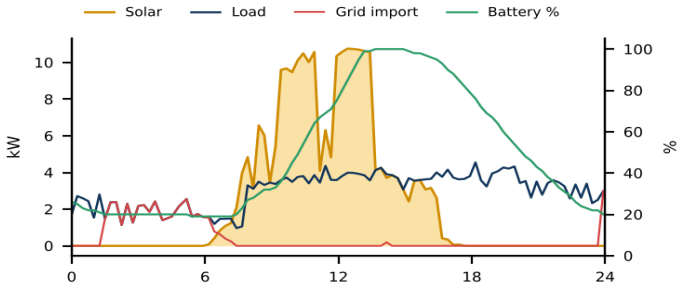
• Correct the PV capacity in SolisCloud. No other action.

Home S1, Songkhla

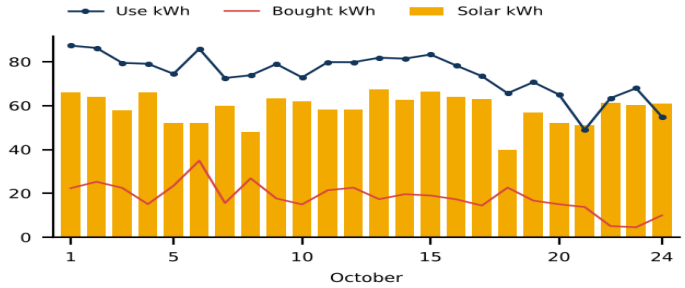
Best yield · check kWp

10 kWp · 10 kW 1-phase hybrid inverter · battery

1414	5.89	75%	18.3	0	24.0
kWh solar	kWh/kWp/day	solar share of use	kWh/day bought	kWh exported	kWh/day from battery



24 October, 5-minute data



Daily totals, 1–24 October

What we found

• Best yield in the fleet: 5.9 kWh/kWp/day. Peak 10.8 kW on '10 kWp' – the registered capacity is probably lower than the panels installed.
• Battery full by 13:10; 8 low-battery and 8 under-voltage alarms.

What to do

• Correct the PV capacity in SolisCloud.
• A second battery would use the afternoon surplus (battery full by 13:10).

How this review was made

- Source: SolisCloud monitoring for all plants Solar ESS installed and monitors there, read on 25 October 2026 (read only – no settings were changed).
- Daily energy for 1–24 October (25 October excluded as incomplete); 5-minute power and battery profiles for 24 October.
- kWh/kWp/day = solar energy ÷ registered PV capacity ÷ days with data. Solar share of use = (use − grid purchase) ÷ use.
- 'Exported unpaid' assumes no PEA sell-back contract.
- Site names, owners, serial numbers and exact addresses are removed; each plant keeps the same label in future monthly reports.
- Gains are estimates and are confirmed on site before any work. Solar ESS supplies energy from the solar system only and is not responsible for the electrical system inside customers' buildings.